UNIVERSITY OF LIMERICK
OLLSCOIL LUIMNIGH

ALGORITHMIC SOUNDSCAPE PERFOMANCE

ASSIGNMENT 2 CS6042

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Overview

The Concept

I created an interactive performance using Freesound Explorer to create the environment of a football match. The performance is controlled by a dice , which acts as the decision maker for every move that is going to happen next.
Every 15 seconds, I roll the die to determine my next action on the sound map. This ensures that the match is driven by a strict algorithmic set of rules rather than my own personal preference, turning the software into a generative system.

Software Used

I chose Freesound Explorer because it doesn't just list sounds it spreads them out on a map like a big field. It puts sounds that look same into groups (clusters). This was perfect for my project because I could treat the map like a real football pitch. I used the quiet areas for players warming up and the loud, crowded areas at the top for the fans. The software lets me zoom in, drag the map, and click dots to play sounds instantly. This allowed me to move around the "stadium" in real time based on the decision of my dice rolls.

Rules

Dice Rule Sheet

- Roll 1 **(Defense):** Click sound for (Jogging/Footsteps).
- Roll 2 **(Possession):** Click 2 sounds at once (Running + Kick).
- Roll 3 **(Long Ball):** Click 1 kick sound
- Roll 4 **(Referee):** Click 1 sharp Whistle and stop moving for 2 seconds.
- Roll 5 **(Fast Attack):** Click 2 sounds very fast in different spots (Sprint + Shot).
- Roll 6 **(Goal):** Click sound for shoot and simultaneously Click 3+ sounds at once (Crowd Roar).

Phase 01: The Kick Off

- **Focus:** Defensive play and finding a rhythm.
- **The Rule:** Roll the die every 15 seconds. If you roll a 1, 2, or 4, perform the action and let the sound breathe. If you roll a 6 early, treat it as a surprise goal and immediately return to defense.
- **Why:** This simulates the slow, tactical start of a match where players are warming up and defending their zone.

Phase 02: Midfield Pressure

Focus: High energy and moving the ball.

The Rule: Focus on Rolls 2, 3, and 5. in between if you roll any other number you can switch to that part i.e if you get a 4 you simply click the whistle sound once give a pause for 2 seconds and start again

Why: This layers the sounds of running and kicking to create a dense, "busy" soundscape as the game intensifies.

Rules

Phase 03: The Final Attack
Focus: Maximum intensity and the final whistle.
Action: Move toward the bottom of the map (Crowd/Atmosphere clusters).
The Rule: Roll more frequently. If you hit a 6 (Goal), perform the "Shoot + Roar" overlay and keep clicking in a Circular Motion to sustain the noise.

The End: Ignore the dice and perform Roll 4 (Whistle) three times, then let go of the mouse for 10 seconds of silence.
Why: This provides a clear climax to the performance, ending with the energetic roar of the fans and the ending with the whistle.

Note for the user

Roll the dice every 15 seconds to determine the next action
Only if you get a 4 then you stop the game for 2 seconds and resume with the next role of the dice
Try to maintain the continous nature of the algorithm

Inspiration(Rationale)

Why I did this?

I became fascinated by how digital maps can be transformed into high pressure environments. For this project, I chose to recreate the sonic tension of a football match to explore the relationship between human instinct and structured algorithms.
I named this performance "The Digital Pitch" because I treated the interface like an arena. Unlike a traditional song, the audio here followed a strict dice based algorithm that acted as my Referee. Every move was a reaction to a roll of the dice, forcing me to manually trigger long balls
The performance was a physical challenge of speed and timing. If my hand stopped moving, the match died. By running through kicks and stacking crowd roars, I had to navigate the map's geography to keep the game's energy alive.
The final result is a three minute journey where sport is performed through chance, and manual interaction based on a set of rules

Limitations:

One of the most challenging yet inspiring limitations was the software's unpredictability. Many sound clusters didn't initially resemble what I expected, which forced me to be incredibly selective in the map and carefully find the right kicks and whistles for my match.
Additionally, many high quality samples only lasted 2-3 seconds. This became a major hurdle because a football match needs constant, flowing energy. To overcome this, I had to physically navigate the map with intense speed, clicking clusters to keep the atmosphere alive. If I slowed down, the stadium went silent.
This difficulty actually improved the performance , the struggle to move the mouse across the screen and react to the algorithm perfectly mimicked the physical exhaustion and fast paced pressure of a real game. These technical constraints turned a simple map into a high stakes instrument that required constant human effort to maintain.

Inspiration(Rationale)

My project is inspired by Manfrod Mohr he used computer algorithms to turn complex geometry into art, and I applied that same logic to my football match. While Mohr used math to decide where lines should go on a grid, I used a dice based algorithm to decide where my mouse should go on the sound map.
I followed the dice rules to navigate the screen, turning a technical interface into a high pressure, unpredictable sporting event.

Conclusion:

The biggest thing I learnt is that limitations can actually make you more creative. At first, I thought the short 2 second sounds and the messy map was a problem. However, I learnt that by using a dice based system, I could turn those problems into a challenge.
I learnt how to work with the software instead of just clicking buttons. Because I had to follow the dice, I couldn't just pick the easy sounds , I had to move fast and react quickly to whatever number I rolled also seeing the clock every 15 seconds . This taught me that a digital tool can feel like a real instrument if you have a strict set of rules to follow. Most importantly, I learnt that the struggle to keep the sounds playing is what actually made the football match feel alive.